

Replication Upgrade Support Procedure for L3 DBA Team

Document Type: Operational Procedure

Purpose: To define the standard process that L3 Database Administrators must follow when the Development team performs application upgrades requiring replication changes.

1. Introduction

During application upgrade cycles, certain changes may require modification of SQL Server Replication components, including adding or dropping articles, validating replication topology, and ensuring data synchronization integrity.

This document outlines the mandatory steps an L3 DBA must perform before, during, and after such changes to ensure reliable and consistent replication operations.

2. Scope

This procedure applies to all SQL Server instances participating in:

- **Transactional Replication**
- **1:1 and 1:Many Replication Topologies**

It covers:

- Pre-upgrade preparation
 - Article management
 - Backup strategy
 - Monitoring and validation
 - Post-deployment tasks
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3. Pre-Deployment Activities

3.1 Index Assessment

- Retrieve and document all **indexes associated** with replication articles from **Subscriber(s)**.
- Ensure that index definitions are available for re-creation if articles are dropped and re-added as part of the upgrade.

3.2 Backup Strategy

Perform backups on the **Published DB(s)** prior to plan downtime:

1. **Differential or Log Backup** based on the maintenance window.
2. Ensure that the backup is valid and restorable, enabling rollback in case deployment fails.
3. Prefer **COPY_ONLY** backups to avoid affecting the backup chain.

Remarks: *Proper timing has not been mentioned. DBA have been allotted only the 30 minutes time window for the DBA-related activities.*

3.3 Article Validation

If articles are expected to be dropped or added:

- List Articles Involved in the Current Deployment. L3 DBA must ensure all articles are available at least one day prior.
- Review / Double validate articles definitions from the **Replication configuration**.
- Prepare to update schedules associated with Dropped/Add operations prior.

Remarks: *As aligned with Sarawut, 2-3 working days before the activity, the list of articles together with the Change Request must have been give already by the Maintenance Team. This will give the L3 DBA Team to plan accordingly, including any needed validation ahead of time.*

4. Deployment Activities

4.1 Article Management

- When removing articles, ensure replication agents (Log Reader Agent and Distribution Agent) are stopped gracefully.
- After changes, re-add articles as per the approved deployment request.
- Recreate all previously documented **indexes after** readding articles.

Remarks: *Specify the replication agents to be stopped (Log Reader / Distribution or both)*

4.2 Replication Health Monitoring

During deployment:

- Monitor the **Published DB(s) log file growth** from deployment **start** until **~2 hours post-completion**.
- Validate that Log Reader Agent and Distribution Agent are functioning normally and have no latency spikes.

4.3 Data Consistency

- Once deployment tasks complete, validate row counts (if applicable) or perform sample checks to confirm data is flowing correctly through replication.

5. Post-Deployment Activities

5.1 Space Recovery

- Post-deployment, review storage usage and reclaim space if necessary (e.g., clean-up of backup files taken on this process).

5.2 Replication Topology Validation

- Confirm the replication setup is healthy in all participating nodes:
 - 1:1 Replication
 - 1:Many Replication
- Verify synchronization status, agent performance, and subscription status.

5.3 Documentation and Reporting

- Update replication configuration records to reflect article changes and index updates. - Where can we find the configuration records? Isn't this should be coming from the Maintenance Team?
- Provide a summary report to Dev and Operations teams, including:
 - Observed issues, if any
 - Final replication health status, With record validations from Publisher / Subscriber(s) (all Articles)

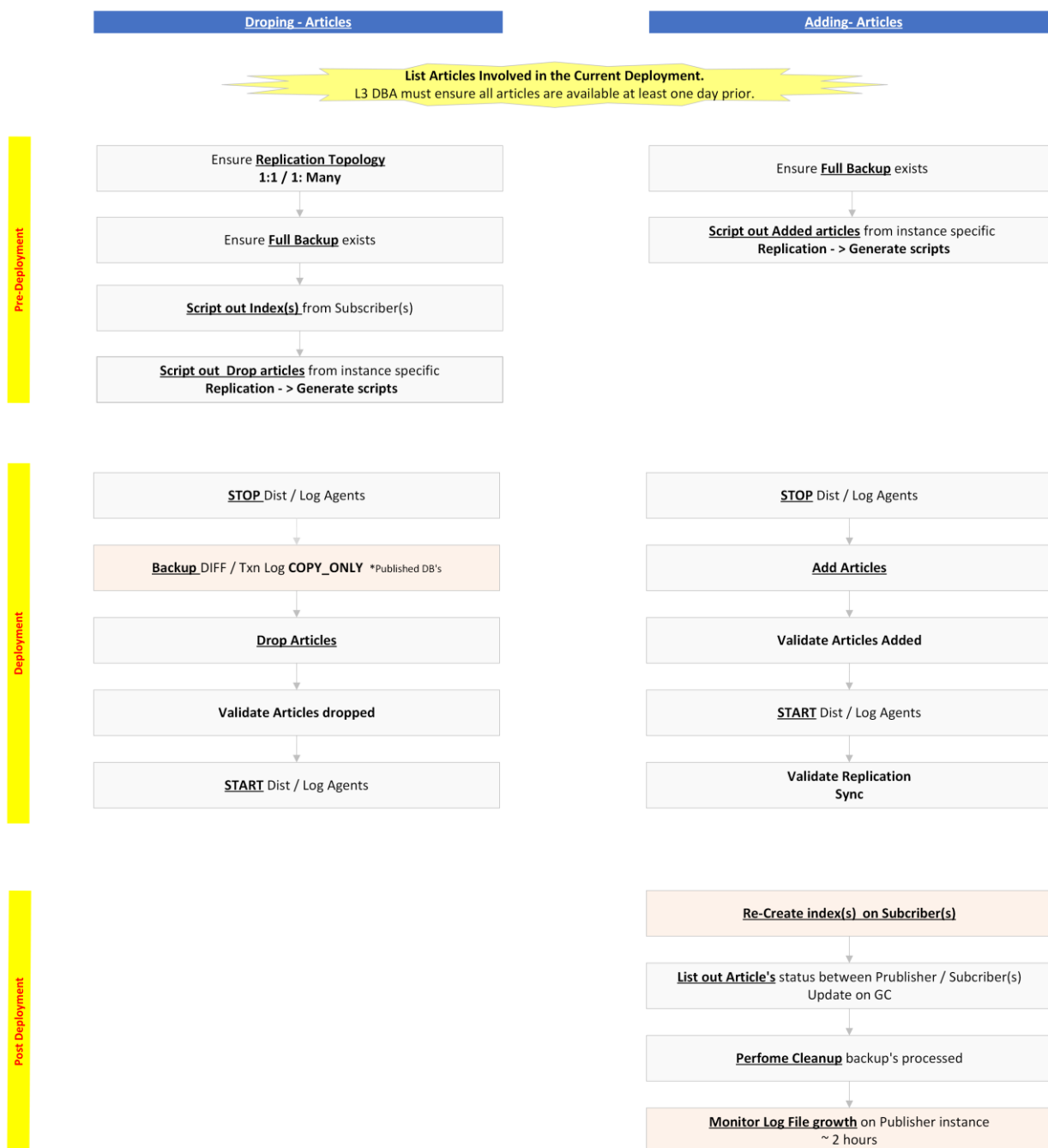
6. Responsibilities

Role	Responsibility
L3 DBA	Executes all steps in this SOP and provides technical validation
Dev oops	Communicates article changes and upgrade requirements

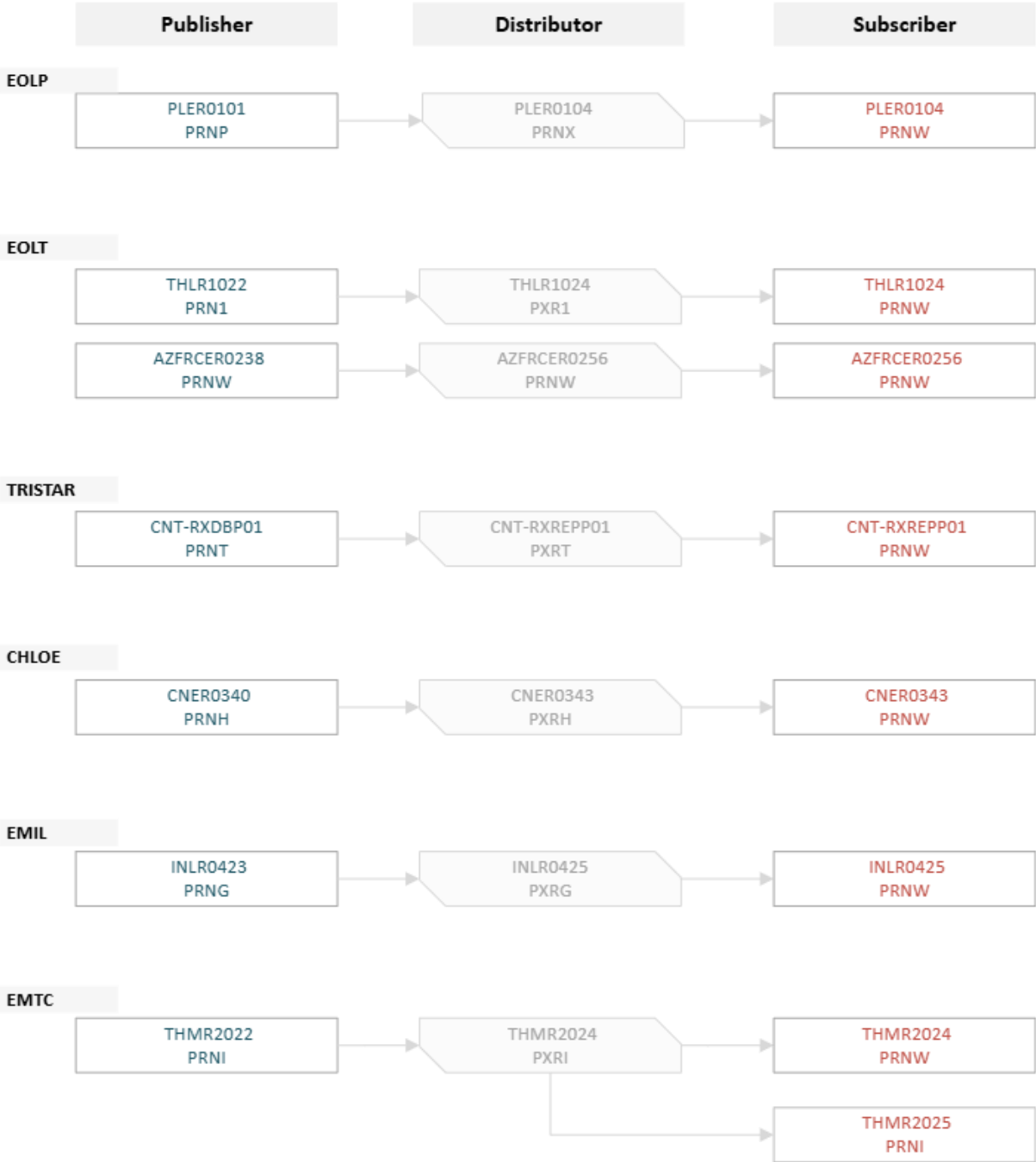
7. Conclusion

Following this procedure ensures controlled replication adjustments, safeguards data integrity during application upgrades, and maintains consistent synchronization across all replicated nodes. L3 DBAs must adhere to this process for every deployment cycle that affects replication systems.

Dropping /Adding Articles



EMEA - Replications (Transactional)



Remarks: Tristar is not included in the EMEA scope.